CRISPR-CAS9 GENE EDITING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on CRISPR-Cas9 Gene Editing

Question 1

10.0 points

Examine the diagram below that illustrates the Cas9 protein–sgRNA complex docking onto target DNA. Describe the molecular interactions between the sgRNA and the target DNA that are critical for specificity. Identify any key residues or structures in the Cas9 protein involved.

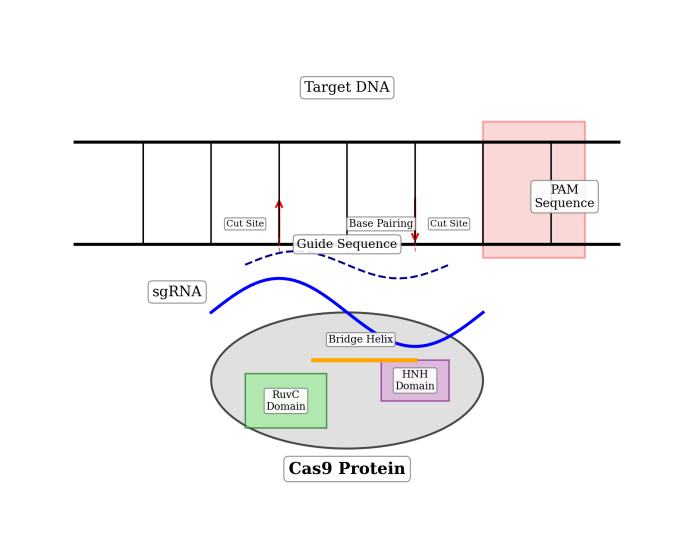

Fig. 1. Diagram of the Cas9 protein–sgRNA complex docking onto target DNA, highlighting the molecular interactions between sgRNA and target DNA. Include key residues or structures in the Cas9 protein that are involved in specificity. Show the spatial arrangement of sgRNA, Cas9, and DNA, with labels for critical interaction points and residues.

Question 2

10.0 points

Refer to the diagram below illustrating the PAM sequence recognition and DNA unwinding mechanism in CRISPR-Cas9. Discuss the role of the PAM sequence in Cas9 activation and how DNA unwinding is achieved.

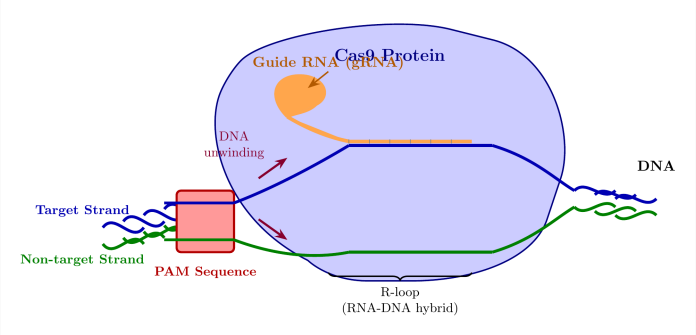

Fig. 2. A diagram illustrating the CRISPR-Cas9 mechanism, focusing on the PAM sequence recognition and DNA unwinding. Include the Cas9 protein complex bound to a DNA double helix, with the PAM sequence highlighted. Show the guide RNA (gRNA) within the Cas9 complex, aligning with the target DNA sequence. Indicate the unwinding of the DNA strands at the target site, with arrows showing the direction of unwinding. Label the PAM sequence, Cas9 protein, gRNA, and DNA strands clearly. IMPORTANT: The diagram is missing required labels. The diagram is missing required labels for the DNA. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram is missing the required labels for the ". Ensure ALL structural components mentioned in the description are clearly labeled.

Question 3

10.0 points

Analyze the diagram below that illustrates DNA double-strand break repair pathways: non-homologous end joining (NHEJ) versus homology-directed repair (HDR). Compare these pathways in terms of accuracy and potential applications in gene editing.

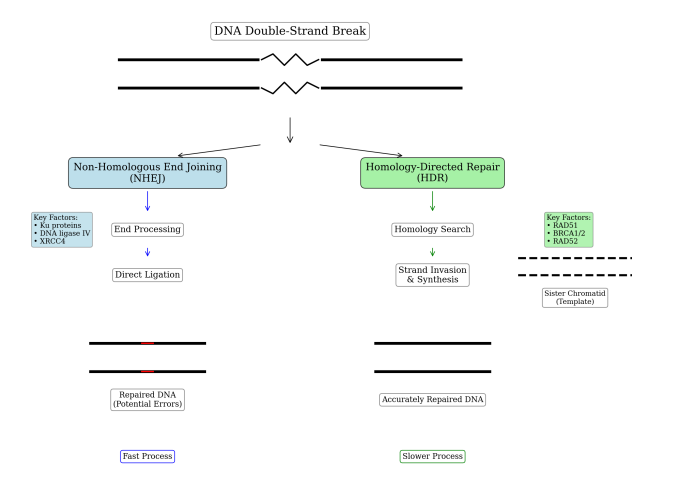

Fig. 3. A diagram illustrating DNA double-strand break repair pathways: non-homologous end joining (NHEJ) and homology-directed repair (HDR). The diagram should show a DNA double-strand break, with two separate pathways branching from it. For NHEJ, depict the direct ligation of broken ends without a homologous template, highlighting potential for errors. For HDR, show the use of a homologous template for accurate repair, indicating the steps of strand invasion, DNA synthesis, and ligation. Label each pathway and include annotations for key steps and differences in accuracy.

Question 4

10.0 points

Analyze the mismatch tolerance depicted in the diagram below for predicting off-target effects in CRISPR-Cas9 gene editing. How do different types of mismatches influence the likelihood of off-target effects?

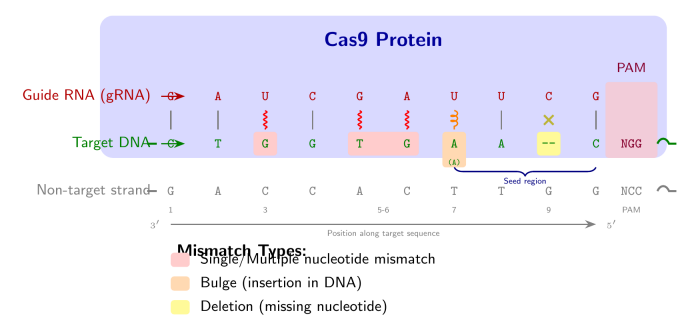

Fig. 4. A diagram illustrating the CRISPR-Cas9 complex with a guide RNA (gRNA) bound to a target DNA sequence. Highlight potential mismatch sites along the DNA where the gRNA may not perfectly align. Show different types of mismatches, such as single nucleotide mismatches, bulges, and deletions, at various positions along the DNA. Label the gRNA, target DNA, and Cas9 protein. Use arrows or markers to indicate the mismatch sites and their potential effects on off-target activity.

Question 5

10.0 points

Examine the base editing window for cytosine base editors (CBEs) in the diagram below. Describe the CBE's base editing window and discuss the implications for specificity and efficiency in gene editing projects.

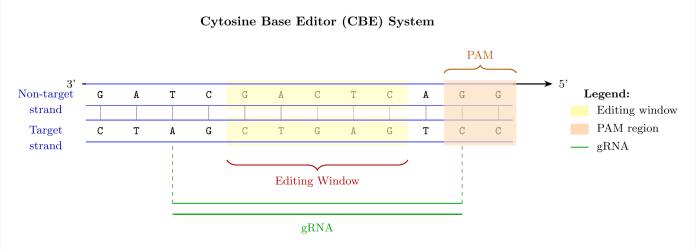

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): The diagram contains an annotation that directly explains; The diagram contains a label "Target C bases" with. A diagram illustrating the base editing window for cytosine base editors (CBEs). Show a segment of DNA with nucleotide bases labeled (A, T, C, G). Highlight the specific region where cytosine to thymine conversions occur, typically a few nucleotides long. Indicate the direction of the DNA strand and label the editing window. Include annotations for the PAM sequence if relevant, and any guide RNA (gRNA) binding sites. The diagram should clearly show the spatial arrangement of the bases and the editing window. IMPORTANT correction needed: The diagram contains an annotation that directly explains. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT correction needed: The diagram contains a label "Target C bases" with. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.